

WEEKLY REPORT - GROUP 02

1. General Information

Field	Information
Group ID	Group 02
Group Name	Group 02
Project Name	Seminar T10: Mutation Testing and Test Effectiveness for the e-Shop SUT
Reporting period	2026-06-29 to 2026-07-04 (Monday to Saturday)
Members	23127216 - Lê Hoàng Lâm (M1); 23127543 - Vu Van Vu (M2)

2. Tasks Completed This Week

Progress below was checked against [seminar/02-planning/weekly-checklist.md](#) and `git log --graph refs/heads/seminar --stat` for 2026-06-29 through 2026-07-04. A task is marked completed only when the [seminar](#) branch contains implementation or measured execution evidence. Drafts and unexecuted proposals are marked partial.

Full Git evidence: [Appendix A - commit_log.txt](#).

23127216 - Lê Hoàng Lâm (M1)

- **[Completed] T4 - Run the full mutation baseline and record the results.**
 - Configured and stabilized the Jest/Stryker test harness for sequential execution.
 - Measured 541 mutants: 175 killed, 73 survived, and 293 with no coverage.
 - Recorded an overall mutation score of 32.35%, together with coverage and per-route analysis, in [seminar/01-research/BASELINE.md](#).
 - Evidence: commits [5fd8f92](#), [9a35d54](#), and [0b53f94](#).
- **[Completed] T5a - Analyze Group A survivors for authentication and coupons.**
 - Extracted structured survivor data from the Stryker HTML report.
 - Classified mutant 28, mutants 397/398, and mutant 396 using root-cause and propagation-level dimensions.
 - Evidence: [apps/backend/scripts/extract-mutants.js](#), [seminar/01-research/survivors.json](#), and commits [53d0161](#), [493f808](#), and [e0c85c7](#).
- **[Partial] T6a - Add kill assertions for Group A survivors.**
 - Designed deterministic clock handling for the authentication boundary mutant and value-pinning assertions for the coupon formula bug, documented in [BASELINE.md](#).
 - Re-checked against current HEAD: [auth.test.js](#) and [coupon.test.js](#) do not yet contain these assertions. Commit [e0c85c7](#) modified only [BASELINE.md](#).

- Evidence: `seminar/01-research/BASELINE.md` (design only); no corresponding test-file diff exists yet.
- **[Partial] T7a - Apply the AI assertion-synthesis validation gate.**
 - `BASELINE.md` records a PASS-27/27-on-original and FAIL-on-mutant (Killed) result for mutants 28, 396, and 397/398, but this cannot be reproduced from current `HEAD`: the underlying assertions are not present in the test files (see T6a). The validation gate is documented, not executable yet.
 - Preserved the AI classification input in `seminar/01-research/clips/2026-07-03_gemini_survivor-classification.md`.
 - Evidence: `seminar/01-research/BASELINE.md` (unverifiable against `HEAD`) and commit `493f808` (classification data only, no test execution).
- **[Partial] Failure-mode handoff and review of Group B.**
 - Technical risks such as SQLite worker races and time-dependent boundary tests were observed, but the required Stryker/AI workflow failure-mode handoff for User Guide section 6 has not been finalized.
 - No current-branch evidence shows completion of the review of T5b/T7b.

23127543 - Vu Van Vu (M2)

- **[Completed] Maintain the FR-08/FR-10 baseline tests used by Cluster B.**
 - Added Supertest baseline coverage for cart/checkout and order-status behavior.
 - Evidence: `apps/backend/__tests__/cart.test.js`, `apps/backend/__tests__/order-status.test.js`, and commit `d1d0904`.
- **[Completed] T5b - Classify and select Group B survivors.**
 - Classified mutant 515 (order-status `canceled to delivered` transition gap) and mutant 268 (cart sequential-state gap) using the same root-cause and propagation-level taxonomy applied to Group A.
 - Triaged mutants 535/536/537 (`require.main === module` startup guard) as Equivalent Mutant / Dead Code candidates, flagged for reviewer sign-off before exclusion from the denominator.
 - Evidence: `seminar/01-research/BASELINE.md` and commit `7e3d4dc`.
- **[Partial] T6b - Write and validate kill assertions.**
 - `BASELINE.md` documents AI-synthesized kill assertions for mutant 515 and mutant 268, but `order-status.test.js` / `cart.test.js` on current `HEAD` do not contain them, and no mutation-score increase was measured.
 - The Validation Gate is explicitly recorded as Blocked in `BASELINE.md` - an honest status, unlike the Group A T6a claim (see Issues).
 - Evidence: `seminar/01-research/BASELINE.md` (design only); no test-file diff.
- **[Partial] T7b - AI assertion synthesis and validation gate.**

- Kill-assertion drafts and a Blocked Validation Gate are documented for mutant 515 and mutant 268.
 - No executed PASS-on-original / FAIL-on-mutant result exists yet; requires adding the assertions and rerunning `npm test / npm run stryker`.
 - Evidence: `seminar/01-research/BASELINE.md` and commit `7e3d4dc`.
- **[Partial] T8 - Metrics and equivalent-mutant triage.**
 - Equivalent-mutant triage completed for mutants 535/536/537 (startup-guard dead code), pending reviewer sign-off before exclusion from the denominator.
 - Setup time, run time, and mutation-score gain per AI-generated test were not measured; the T8 metrics table in `BASELINE.md` reuses Lâm's T4 full-run numbers instead of an independent measurement.
 - Evidence: `seminar/01-research/BASELINE.md` and commit `7e3d4dc`.
- **[Partial] Failure-mode contribution and review of Group A.**
 - A state-machine test-gap note was drafted, but it has not been integrated into the current branch's User Guide failure-mode section.
 - No current `seminar` branch commit records Vũ's review of the completed Group A analysis in `e0c85c7`.

Consolidated Evidence

Commit	Date	Owner	Evidence
<code>d1d0904</code>	2026-06-30	Vũ	Added FR-08 cart and FR-10 order-status Supertest baselines
<code>5fd8f92</code>	2026-06-30	Lâm	Forced sequential Jest workers to remove SQLite race failures
<code>9a35d54</code>	2026-06-30	Lâm	Stabilized the backend test harness
<code>0b53f94</code>	2026-06-30	Lâm	Recorded measured coverage and mutation baseline
<code>341a94e</code>	2026-07-03	Lâm	Added the generated Stryker baseline HTML report
<code>53d0161</code>	2026-07-03	Lâm	Added the Stryker report extraction script
<code>493f808</code>	2026-07-03	Lâm	Added survivor data and AI classification evidence
<code>e0c85c7</code>	2026-07-03	Lâm	Documented Group A T5a survivor classification and proposed (not yet implemented) T6a/T7a kill assertions

Commit	Date	Owner	Evidence
7e3d4dc	2026-07-04	Vũ	Documented Group B T5b survivor classification (mutants 515, 268) and equivalent-mutant triage (535/536/537); kill assertions and metrics not yet executed
6a4ee97	2026-07-04	Lâm	Normalized survivor-entry formatting between Group A and Group B in BASELINE.md (no content change)

Appendix A - Raw Git Commit Log

- Evidence file: [commit_log.txt](#)
- Scope: branch **seminar** only, 2026-06-29 00:00:00 through 2026-07-04 23:59:59, timezone **+07:00**.
- Command: `git log --graph refs/heads/seminar --stat --since=2026-06-29T00:00:00+07:00 --until=2026-07-04T23:59:59+07:00 --date=iso-strict --pretty=fuller`

3. AI Usage Declaration

AI Usage Notes

Gemini Version not recorded by the tool, Google, [gemini.google.com](#) (chat link: <https://gemini.google.com/app/0aec0f27e424e1ff>), accessed 2026-07-03 (exact hour not recorded in the saved clip).

Prompt:

"Theo các kết quả nghiên cứu về Mutation Testing, hãy giúp tôi tìm kiếm và tổng hợp các loại survivor có thể có khi thực hiện Mutation Testing"

Used to source a two-axis taxonomy for classifying Stryker survivors: Root-Cause Classification (Equivalent Mutant / Assertion Gap / Boundary Value Blindness / Dead Code) and Propagation Level (No Coverage / Weakly Survived / Strongly Survived). AI generated the generic taxonomy and terminology, preserved verbatim in [seminar/01-research/clips/2026-07-03_gemini_survivor-classification.md](#). M1 mapped it onto the actual mutants (28, 396, 397/398 for Group A; 515, 268 for Group B), added an extended label the AI did not produce ("Missing Test Case"), and recorded the mapping in [BASELINE.md](#).

4. Tasks Planned for Next Week

The next planning period is 2026-07-06 through 2026-07-12.

Lê Hoàng Lâm (M1)

- **[F2]** Write User Guide section 3, an end-to-end first-test walkthrough for the coupon feature in no more than 15 steps.
- **[F4]** Write User Guide section 6 and consolidate at least three verified Stryker/AI workflow failure modes.
- Review Vũ's F1 and F3 sections before the User Guide is submitted.
- **[T9]** Build the "Kill the Mutant" sandbox with five checkout/coupon mutants and a scoring flow that runs within four minutes.

- [D1] Draft the seminar demo script, covering both Stryker and the AI feature within ten minutes.

Vu Van Vu (M2)

- [F1] Write User Guide sections 1 and 2: Introduction and Installation, with exact commands and at least one screenshot.
- [F3] Write User Guide sections 4, 5, and 7: Advanced Usage, Troubleshooting with at least three real errors and fixes, and References.
- Review Lâm's F2 and F4 sections.
- [E1] Create `Activity_Worksheet.md` and its answer key, reproducible within 25 minutes without group assistance.
- [G1] Start `Seminar_Slides.pptx` with no more than 15 slides following the approved outline.

Shared

- By 2026-07-08, merge and review all seven User Guide sections, confirm at least three evidence-backed failure modes, and verify every reference.
- Before starting the User Guide, finish the unverified execution evidence for both groups - Group A T6a/T7a and Group B T6b/T7b/T8 - so the guide does not rely on candidate or placeholder data.

5. Issues

Issue	Impact	Required action
Group B T6b/T7b lacks PASS-original and FAIL-mutant execution evidence.	The selected assertions cannot yet be claimed to kill real mutants.	Add assertions to the owned test file, run Jest and Stryker, and record mutant IDs and results.
Group A T6a/T7a were documented in <code>BASELINE.md</code> as implemented and validated, but commit <code>e0c85c7</code> only modified <code>BASELINE.md</code> ; <code>auth.test.js/coupon.test.js</code> do not contain the described assertions.	The Completed status previously recorded for T6a/T7a was not backed by an executable diff, the same gap as Group B.	M1 adds the assertions to the test files, reruns <code>npm test / npm run stryker</code> , and replaces documented results with real output before claiming Completed.
T8 has completed equivalent-mutant triage but no measured timing/score-delta metrics.	The 05/07 gate is not fully satisfied.	Capture setup time, run time, and score delta per AI test; triage results are already recorded.